

SPECIFICATION OF ACOUSTIC PERFORMANCE OF PARTITIONS

1.0 Scope

- 1.1 This specification relates to the acoustic performance of partition.
- 1.2 This specification is addressed to the Contractor who shall comply with its requirements.

2.0 Acoustic Specification

2.1 Acoustic Standards

The Sound Transmission Class and Noise Isolation Class are defined according to ASTM E90. Noise reduction measurements will be made in general conformance with ASTM E336-84 or the latest version as it applies to measurement of isolation between rooms. The NIC rating will be calculated in accordance with ASTM E413.

2.2 Sound Insulation Requirements

1. The partition shall provide the following minimum sound insulation as follows:

Acoustical Performance			
Separation / Space	STC Rating ⁽¹⁾ (Laboratory)	NIC Rating ⁽²⁾ (Field)	Partition Type
*Office type premises: Security Control Room & Management Offices / Network Operation Centre /Office	50	43	B1/B2
*Conference Rooms	55	48	A1/A2
*Between Mechanical Equipment Rooms and Office Services Corridor	55	48	A1/A2
*Between Mechanical Equipment Rooms and Occupied Rooms	65	58	F
Internal glass window for Network Operation Centre	40	-	G
Corridor glazed wall for Conference Rooms	49	42	H

⁽¹⁾ Sound Transmission Class

⁽²⁾ Noise Isolation Class

Tested nominally at 1.5m from each side of partition.

(*) In accordance to the requirements of HKBEAM Plus and LEED.

2. The partition type as described in the above section should have construction as follows:

Partition	STC Rating	Construction
Type A1	STC 55 R.C. wall	150mm thick R.C. concrete wall (density of 2400kg/m ³) with minimum surface weight of 360kg/m ² , erected from slab to slab.
Type A2	STC 55 Solid Concrete Block Wall	150mm thick solid concrete block (density of 2000kg/m ³) and 15mm thick plastering on both sides with minimum surface weight of 360kg/m ² , erected from slab to slab.
Type B1	STC 50 R.C. wall	100mm thick R.C. concrete wall (density of 2400kg/m ³) with 10mm thick plastering on both sides, minimum surface weight of 280kg/m ² , erected from slab to slab.

Partition	STC Rating	Construction
Type B2	STC 50 Solid Concrete Block Wall	100mm thick solid concrete block (density of 2000kg/m ³) and 20mm thick plastering on both sides with minimum surface weight of 280kg/m ² , erected from slab to slab.
Type F	STC 65 Furred Drywall Partition	Applying 50mm metal stud with 50mm insulation (density of 48 kg/m ³) and double layer of 15mm thick gypsum board, erected from slab to slab, at the 150mm thick solid concrete block wall. This furred partition shall be supported on ceiling and floor slabs without touching the wall. The furred partition shall either be built having a 40mm gap without direct contact with the concrete wall or mounting the metal studs on resilient mount if necessary (similar to Kinetics IsoMax or equivalent).
Type G	STC 40 Glazed Window	Double laminated glazed window with construction of "8mm glass+12mm air space+12mm laminated glass" or Laminated single glazed window with construction of "8mm glass+1.52mm PVB+12mm glass", or similar to achieve the required acoustic rating. The glazing shall be fitted in a solid framework and all gaps along the junction shall be sealed up properly by sealant to block any sound leakage. Ceiling baffle above glazed panel shall be Type B1/B2 partition.
Type H	STC 49 Glazed Wall	Laminated double glazed demountable partition with construction of "13.52 mm thick laminated glass + about 75 mm air gap + 11.52 mm thick laminated glass" or similar to achieve the required acoustic rating (e.g. "JEB X-Series", "Clestra M1" or equivalent). The glazing shall be fitted in a solid framework and all gaps along the junction shall be sealed up properly by sealant to block any sound leakage. Ceiling baffle above glazed panel shall be Type B1/B2 partition.

- 2.3 Perimeter of the partitions shall be constructed so that they provide sound insulation similar to that of the partition.
- 2.4 Gaps between partition and surrounding structure shall be as small as practicable. Fill gaps greater than 6 mm with dense filler to a depth of not less than 50 mm. Seal both surfaces of the joint between the partition and structure with a non-hardening sealant.
- 2.5 All fixing and jointing details shall be submitted to the Project Manager for approval.
- 2.6 Any electrical and service outlets for adjacent rooms shall be positioned minimum 150mm apart ideally. All the wall thickness between any back-to-back sockets shall maintained minimum 100mm.
- 2.7 For walls with TV panel, inappropriate fixing detail which creates large opening on the wall partition shall avoid in order to maintain the acoustic integrity.
- 2.8 Acoustic test report shall be submitted to the Project Manager for approval.
- 2.9 Use adequate numbers of skilled workmen who are thoroughly trained and experienced in the necessary crafts and who are completely familiar with the specified requirements and the methods needed for proper performance of the work of this section.

- 2.10 The Contractor shall be on hand to observe the field testing and to make partition adjustments as required. Cost of the field tests will be the responsibility of the Contractor. Field tests shall be conducted in accordance with E336 for Noise Reduction and the Noise Insulation Class (NIC) rating will be calculated in accordance with ASTM E413. The partitions for the concerned rooms as mentioned above shall be tested to ensure the NIC ratings.
- 2.11 The NIC rating may be adjusted in accordance to the actual site and measuring conditions, for example, the ratio of door to front partition will need to be taken into consideration because of their difference in NIC ratings. This adjustment shall be determined case by case and verified by the project acoustic consultant.
- 2.12 Where any failure or non-compliance of the acoustic integrity is attributed to the work of this trade, all the remedial work shall be carried out by the Contractor until the required acoustic properties are achieved. Any re-testing required due to product installation not meeting the acoustical performance will be the responsibility of the Contractor. Test proposal shall be submitted to Project Manager for approval prior to test.

END OF SECTION